Retrack: An Occlusion Aware Multi Camera Multi Person Tracking System

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Abstract

Multi-camera person tracking is an important task in intelligent surveillance and monitoring systems. However, maintaining accurate tracking remains challenging due to frequent occlusions and the need for reliable person re-identification when individuals are temporarily blocked from view or move across different camera views. Our project proposes a real-time multi-camera person detection and tracking system that identifies and tracks individuals in live video streams. The proposed system focuses on improving robustness to occlusions and person re-identification.

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1. Introduction

Retrack is a final year project that provides a web application for multi-camera pedestrian detection and tracking in a calibrated scene. It combines synchronized images from seven fixed cameras into a shared bird eye view, estimates pedestrian ground positions and associates' observations over time. Approved operators can configure monitoring, view camera and ground-plane results, and retrieve reproducible session reports.

Multi-View Object Detection. Tracking objects across many cameras is a difficult task since objects can appear very different from different angles, be obscured by other objects, or be visible in one camera but not in others. Multi-Target Multi-Camera tracking has significant issues in dense areas, as it faces occlusion and identity switches. Multi-camera systems with overlapping fields of view can be leveraged to perform enhanced detection and tracking outperforming single-camera systems as they can handle occlusions more robustly.

This Software Requirements Specification defines the project vision, users, scope, functional behavior and quality requirements. It supports implementation by the development team and review by the supervisor. The word shall identify a required behavior. Proposed performance targets require measurement before they can be claimed as achieved.

The research basis is TrackTacular [2], which extends EarlyBird [1], builds on the MVDet [4] multi-view detector and is evaluated on the WILDTRACK dataset [3]. AIModel is the application integration interface that wraps the TrackTacular backend. Feature extraction, calibrated BEV fusion, ground-plane detection and temporal association form the processing pipeline.

The two account roles are End User and Admin. A new End User registers through the public registration page and can sign in after Admin approval. A browser authentication session and a monitoring analysis session have separate lifecycles. Track IDs are unique within one analysis session and do not establish a person's real identity or guarantee continuity through occlusion.

2. Vision Document

2.1 Problem Statement

Table 1 Problem Statement

Category	Description
Problem	Operators must compare overlapping camera views manually to understand pedestrian positions and movement. Research scripts provide model output but lack an accessible account, session and report workflow.
Affects	Approved monitoring operators, administrators controlling access, and academic teams reviewing multi-view detection and tracking.
Impact	Manual comparison and counting take time and can produce inconsistent summaries. Incomplete, mismatched or incorrectly calibrated input can make results misleading.
Solution	Provide a browser workflow for approved accounts, validated synchronized seven-camera input, calibrated ground-plane detections, session-local trajectories, visual monitoring and reproducible reports.

2.2 Business Opportunities

The immediate opportunity is an academic prototype that makes a research tracking pipeline usable without requiring an operator to run inference scripts. Demonstrated capabilities could inform later work for campuses, plazas, transport facilities or other fixed camera installations.

A browser interface and recorded replay can support repeatable evaluation, operator demonstrations and later integration work. Extending the system to another site would require suitable calibration, compatible model settings and evidence from that scene.

2.3 Objectives

- Provide approved End Users with account-controlled configuration, Start, view and Stop operations.
- Accept LIVE and RECORDED sources through the same synchronized seven-camera FrameSet contract.
- Validate calibration, checkpoint compatibility and input metadata before inference.
- Produce pedestrian ground positions and temporal trajectories in a common BEV, with session-local track IDs.
- Display camera annotations and BEV movement information without blocking input processing.
- Store committed processed frame records and generated consistent PDF and CSV reports.
- Provide Admin with account and profile decisions, checkpoint selection and diagnostic records.
- Record reproducible research and application test evidence without treating an offline metric as a web-system acceptance result.

2.4 Scope

The initial product is a desktop web application for the configured WILDTRACK-style seven-camera scene. Its application scope includes End User registration; approved-account login and logout; Admin account decisions; requested and approved profile changes; calibration upload or selection; compatible checkpoint management; LIVE and RECORDED monitoring; input validation; pedestrian detection and tracking; camera and BEV display; persistent results; reporting; and audit records.

Reports are generated automatically when an eligible session is finalized. Authorized users may also generate, search, view and download reports manually. The report contract includes processed frames, session-local unique tracks, mean occupancy and peak occupancy. Failed sessions with committed results produce clearly identified partial reports; empty sessions do not produce reports.

The scope excludes behavior recognition, non-pedestrian object categories, face recognition, real-person identity, cross-session re-identification, camera-calibration estimation, model training through the browser, native mobile applications and arbitrary camera layouts. Offline model training and checkpoint preparation remain development and research activities. The LIVE contract does not assert support for every commercial camera transport protocol.

2.5 Constraints

Table 2 Prototype Constraints

ID	Constraint	Implication
C-01	Exactly seven configured fixed cameras.	Use camera order CVLab1 to CVLab4, then IDIAP1 to IDIAP3; one image is required from each camera.

ID	Constraint	Implication
C-02	Compatible intrinsic and extrinsic calibration for every camera.	Use one shared ground-plane convention and centimeter translation units; camera movement invalidates the corresponding calibration.
C-03	Accepted images are 1920 by 1080 pixels.	Decode each image; use only preprocessing compatible with the selected checkpoint.
C-04	Synchronized source metadata and increasing source order.	Input adapters must preserve common frame index/time; coincidence arrival alone does not prove physical synchronization.
C-05	The scene configuration must match the checkpoint.	Uploading another calibration alone does not establish support for a new site.
C-06	One active analysis session per model instance.	A second Start on a busy instance is refused; report retrieval remains available.
C-07	A compatible trained checkpoint and documented inference environment are prerequisites.	Persistent application records must remain outside an ephemeral GPU notebook runtime.
C-08	CUDA reference host for performance evaluation.	Record hardware and configuration; CPU throughput and production latency are not guaranteed.
C-09	Pedestrians and session-local tracks only.	Do not interpret a track ID as a real identity or uniqueTracks as a guaranteed count of unique real people.

The three-member project team must work within the academic schedule and available GPU resources. Network interruption, unavailable cameras and model/resource failures must produce explicit states. Repository revisions, checkpoint configuration, dataset use and applicable licenses must be recorded.

2.6 Stakeholder and User Description

2.6.1 Market Demographics

The primary audience is the project team, supervisor, academic reviewers and approved operators of a fixed calibrated camera installation. Potential later users include campus operations staff and researchers studying occupancy and pedestrian movement. The prototype is not a general consumer application and does not target every camera layout or public deployment.

2.6.2 User Environment

An End User accesses the application through an authenticated desktop browser with enough screen space for camera views and a BEV panel. The server provides a web service, input adapter, inference worker and persistent metadata/report storage. A live source supplies synchronized frame sets; a recorded source supplies ordered replay and an explicit end-of-source condition.

The team uses a cloud GPU runtime for offline experiments. This is the experimental environment, not a minimum deployment specification. A fresh Colab runtime requires dependencies to be restored; checkpoints and application results need persistent storage. Exact server resources and tested browser versions will be recorded in acceptance evidence.

2.6.3 Stakeholder Profiles

2.6.3.1 Supervisor Team

Table 3 Supervisor Team

Field	Description
Representatives	Dr. Hashim Yasin
Description	Supervisor who guides the scope, design and academic evaluation.
Type	Technical and academic stakeholder.
Responsibility	Review requirements, architecture, research methods, progress and conclusions.
Success Criteria	Committed features and reproducible evidence are consistent with the approved scope.
Involvement	Requirements and design reviewer; progress and evaluation reviewer.
Comments and Issues	None

2.6.3.2 Development Team

Table 4 Development Team

Field	Description
Representatives	Raja Sharjeel Ashraf, Rizwan Mustafa Khan and Abdul Hadi
Description	Three-member team responsible for the FYP prototype and its documentation.
Type	Technical stakeholders.
Responsibility	Implement the web workflow and model integration; manage checkpoints; validate data; maintain documentation and tests.
Success Criteria	A reproducible demonstration and traceable behavior meeting the approved requirements.
Involvement	Research, design, implementation, testing and documentation.
Comments and Issues	None

2.6.3.3 End Users

Table 5 End Users

Field	Description
Representatives	Approved monitoring operators.

Field	Description
Description	People who configure and operate their own analysis sessions and retrieve their reports.
Type	External operational stakeholders.
Responsibility	Supply compatible calibration/source settings, monitor results and use approved account permissions.
Success Criteria	Clear status, usable camera/BEV display and consistent reports for their authorized sessions.
Involvement	Workflow feedback and acceptance demonstrations.
Comments and Issues	Public registration creates a Pending End User account; access requires Admin approval.

2.6.3.4 Academic Reviewers

Table 6 Academic Reviewers

Field	Description
Representatives	FYP evaluation panel and research reviewers.
Description	Reviewers assessing the design, demonstration and experimental claims.
Type	Academic stakeholders.
Responsibility	Assess scope consistency, technical understanding, reproducibility and evidence.
Success Criteria	Research results and web-system tests are clearly separated and supported.
Involvement	Evaluation and technical feedback.
Comments and Issues	Reviewers do not have an application account role.

2.6.3.5 Admin

Table 7 Admin

Field	Description
Representatives	Authorized administrator with a provisioned Admin account.
Description	Privileged user responsible for approvals, configuration and diagnostics.
Type	Internal operational stakeholder.
Responsibility	Decide account/profile requests; select compatible checkpoints; inspect logs and authorized reports.

Field	Description
Success Criteria	Administrative actions are protected, auditable and leave consistent resource states.
Involvement	Administrative workflow review and acceptance testing.
Comments and Issues	Public registration cannot create or promote an Admin; monitoring operations belong to End User.

2.6.4 Stakeholder Summary

Table 8 Stakeholder Summary

Stakeholder	Description	Responsibility
Supervisor Team	Supervisor who guides the scope, design and academic evaluation.	Review requirements, architecture, research methods, progress and conclusions.
Development Team	Three-member team responsible for the FYP prototype and its documentation.	Implement the web workflow and model integration; manage checkpoints; validate data; maintain documentation and tests.
End Users	People who configure and operate their own analysis sessions and retrieve their reports.	Supply compatible calibration/source settings, monitor results and use approved account permissions.
Academic Reviewers	Reviewers assessing the design, demonstration and experimental claims.	Assess scope consistency, technical understanding, reproducibility and evidence.
Admin	Privileged user responsible for approvals, configuration and diagnostics.	Decide account/profile requests; select compatible checkpoints; inspect logs and authorized reports.

Table 9 Role Permissions

Operation	End User	Admin
Submit End User registration	Before authentication	No public Admin creation
Log in and out	Own account	Own account
Decide account or profile requests	No	Yes
Request/apply approved profile edit	Own account	Decision only
Configure/start/stop/view monitoring	Own sessions	Not assigned
Generate/search/view/download reports	Own sessions	Authorized reports
Choose checkpoint and inspect logs	No	Yes

3. Literature Review

Late fusion. Late fusion detects the person(s) in each camera separately and combines results afterwards. In traditional approaches, each camera independently detected pedestrians and then passed the results to the next stages: spatial association to match detections across cameras and temporal association to maintain identities across frames. There were graph-based approaches initially; however, inaccuracies introduced by each camera can propagate into the next stage, and that can reduce tracking accuracy. It is a simpler approach but has some tradeoffs. Firstly, it cannot recover what a camera has missed at the merging stage. Secondly, the nature of the approach means that any issues in individual detections then propagate errors when results are combined.

Early fusion. Early Fusion combines information or learned features from all cameras onto a shared representation such as a bird's-eye-view (BEV) map which allows for joint detection. This is particularly helpful in the case of occlusion where weak cues from multiple views can reinforce each other and help detect individuals that would have otherwise been discarded. The tradeoff with early fusion is that it requires calibrated and synchronized cameras with overlapping fields of view.

MVDet. In [4], the authors follow a form of early fusion whereby instead of projecting final detections, they project each camera's feature map onto the ground plane in a process called feature perspective transformation and then stack the views. They then use large-kernel convolutions to reason over the whole ground plane. They introduced an anchor free approach by producing an occupancy heatmap instead of predicting bounding boxes. This approach had several advantages over existing methods that used Multiview anchor box features as representation which hypothetically caused degraded performance in cases where the pre-defined anchor boxes were inaccurate. The proposed model by [4] is end-to-end learnable and outperformed the state-of-the-art model by 14.1% MODA on Wildtrack dataset. In [4], the authors also presented a new synthetic dataset, MultiviewX which allowed the authors to control the level of occlusion. Furthermore, they performed spatial aggregation through a fully convolutional approach which replaced the need for any design and operations aside from CNN.

Multiview and spatial aggregation. The authors of [4] frame multiview pedestrian detection around two questions. The first is multiview aggregation, which asks how cues from several calibrated cameras should be combined for each location on the ground plane. The second is spatial aggregation, which asks how information from neighbouring ground locations should be combined to resolve the ambiguities caused by crowds and occlusion. Because camera calibrations map each ground point to a pixel in every view, these systems usually evaluate a pedestrian occupancy map (POM) on the ground plane rather than bounding boxes [3, 4].

Earlier aggregation methods. Before [4], multiview systems either fused single-view detection results or pooled features from multiview anchor boxes. As reviewed in [4], late-fusion methods such as RCNN & clustering and POM-CNN project each camera's detections onto the ground, so a person missed in a view cannot be recovered. Anchor-based methods such as DeepMCD and Deep-Occlusion place a box of assumed human height and width at every ground location and pool the image features inside it. The authors of [4] argue that these boxes are often inaccurate, for example for a sitting person, so the pooled features describe the background to a large extent. Spatial reasoning in these methods relied on conditional random fields (CRFs) or mean-field inference, which require hand-designed potential terms and operations outside the CNN.

Feature perspective transformation. MVDet first extracts a C-channel feature map from each view with a shared ResNet-18, replacing the last strided convolutions with dilated convolutions to keep a high spatial resolution. These feature maps are projected onto a common ground plane using perspective transformation based on the camera's intrinsic and extrinsic parameters. While perspective transformation only translates image information onto a flat plane without preserving information about height, it is parameter-free and exact for points on the ground. Assuming every pixel lies on the ground ($z = 0$), the 3

× 4 camera projection matrix reduces to a 3 × 3 matrix, which defines a sampling grid that warps each feature map onto a quantized ground grid, with out-of-view cells filled with zeros. The N projected maps are concatenated with a 2-channel X-Y coordinate map, giving an (N × C + 2)-channel ground-plane feature maps. The authors also tested projecting RGB images and single-view detection results instead of feature maps. Projecting raw images breaks the spatial relationship between neighbouring pixels, while projecting detections limits the information that can be aggregated, so feature maps give the best trade-off.

Spatial aggregation. Instead of CRFs, [4] applies three 3 × 3 convolutional layers with dilations of 1, 2 and 4 on the ground-plane feature map. Each output cell therefore sees a 1.5 m × 1.5 m neighbourhood, which lets the network reason jointly about nearby people while staying fully convolutional and end-to-end trainable. The last layer outputs a single-channel occupancy map.

Training and inference in MVDet. MVDet is trained as a regression problem: the ground-truth occupancy is blurred with a Gaussian kernel into a soft target, and the network output is compared to it with a Euclidean loss. An auxiliary single-view loss supervises head and foot points in every camera, weighted by $\alpha = 1$. At test time the occupancy map is thresholded at 0.4, and non-maximum suppression (NMS) with a 0.5 m radius removes duplicate detections; 0.5 m is also the distance used to count a detection as a true positive in evaluation [3].

Results of MVDet. On Wildtrack, MVDet reaches 88.2% MODA, the primary metric that accounts for both missed detections and false positives [4], compared with 74.1% for Deep-Occlusion as reported in [4]. On MultiviewX it reaches 83.9% MODA. The ablations support each design choice: on Wildtrack, projecting images instead of feature maps drops MODA to 26.8%, projecting single-view results gives 68.2%, and removing the large-kernel convolutions gives 76.9%. Removing the single-view loss costs 1.2 and 2.0 MODA points on Wildtrack and MultiviewX respectively. On MultiviewX, MODA also falls from 89.6% to 69.8% as crowd density rises from 20 to 80 people per frame, showing that heavy occlusion remains difficult.

Limitations of MVDet. Some limitations of [4] include flat-floor projection smearing tall bodies across the map, no tracking and no use of temporal information. Because the head and torso are above the ground, projecting them can place their features away from the person's actual position, creating distortion shadows. As reviewed in [1, 2], later work tried to correct these shadows: MVDeTr uses a shadow transformer with deformable attention, SHOT projects features at several heights with stacked homographies, and MVTT projects regions of interest to the estimated foot positions.

Lifting strategies. There are several families of lifting strategies including: depth-based, attention-based and parameter-free sampling. As described in [2], depth splatting instead estimates the image points' distance from the camera to be able to place this information in three-dimensional space with varying degrees of accuracy depending on the quality of depth estimations. When used in Simple-BEV [6] and other methods, bilinear sampling is performed in the opposite manner, as first location in space is defined and afterwards its projection is derived from the camera images and matching pixel value is sampled. While deformable attention application in BEVFormer allows extracting necessary pixels depending on their usefulness, it is more time-consuming and requires processing big datasets. Finally, bilinear sampling becomes especially useful when multiple cameras are installed in the area of interest, as the configuration of their position allows determining whether the information extracted from the images corresponds to the given three-dimensional location, which is a process that relates to triangulation. According to the findings listed in Simple-BEV [6], training details like batch size and image resolution matter more than lifting choices. The main limitation of [6] is that the findings are derived for self-driving vehicle segmentation which may not fully translate to pedestrian datasets. The results of [2] on Wildtrack largely agree, as perspective transformation, bilinear sampling and depth splatting are within 1.4 MODA of each other. However, deformable attention falls far behind at 78.4 MODA, which

[2] attributes to overfitting on small datasets, so the choice of lifting strategy still needs a closer look on multiview pedestrian datasets.

ByteTrack. Unlike most trackers, ByteTrack [5] does not discard low-confidence detections and instead matches high-score detections first and then uses the low-confidence detections to continue existing tracks. This second matching uses box overlaps (IoU) only, because appearance features of occluded people are unreliable [5]. Low-score detections that match no existing track are discarded as background, so the tracks act as a filter separating occluded people from false detections. This approach helps in recovering occluded people whose detections may otherwise have been thrown away. However, it performed detection and tracking over a single camera only and 2D boxes.

EarlyBird. EarlyBird [1] made significant progress in this field by proposing an early fusion approach for tracking in BEV. EarlyBird is built upon MVDet [4], which uses early fusion to detect pedestrians in a shared BEV. It also takes inspiration from FairMOT which in conjunction learns pedestrian detection and re-identification. To address the problem of distortion shadows, EarlyBird uses a larger receptive field through a ResNet-based BEV decoder. The network outputs person centers and a feature vector describing each person's appearance allowing them to be recognized later. EarlyBird uses two distance measures for temporal association. Cosine distance compares re-ID embeddings to evaluate appearance similarity, while Kalman Filter-based Mahalanobis distance measures how closely a new detection matches with a previous one. If some detections and tracklets remain unmatched then EarlyBird performs a second matching stage using the distance between their BEV centres. Unmatched detections initiate new tracks, while unmatched tracklets are retained for up to 10 timesteps to support identity recovery following temporary occlusions. It out-performs the state-of-the-art tracking methods and improves the current state-of-the-art on Wildtrack by +4.6 MOTA and +5.6 IDF1. Despite showing a significant improvement, the authors of [1] acknowledge that EarlyBird requires accurately calibrated and synchronized cameras along with high-quality 3D annotations, and higher computational resources because features from multiple cameras are processed together.

TrackTacular. Lifting Multi-View Detection and Tracking to the Bird's Eye View [2], referred to by its authors as TrackTacular, is a work by the same authors as [1] that builds upon [1] and compares multiple lifting strategies: perspective transformation [4], depth splatting, bilinear sampling [6] and deformable attention. It also adds temporal history allowing the model to look at the previous frame's BEV. It drops [1]'s explicit re-ID embedding and instead learns motion offsets from past and current BEV features, which the authors argue encode both appearance and motion cues. The findings indicate bilinear sampling as a viable alternative to depth splatting in terms of performance in pedestrian datasets. Despite bilinear sampling being a good alternative to depth splatting, depth splatting performs better on the Synthehicle dataset, mainly when it is tested in scenes that differ from the ones being a part of the training. Moreover, various challenges with deformable attention have been identified in smaller datasets with a high risk of model overfitting. However, in the ablation study of [2], which is reported only on MultiviewX, enabling temporal history helps find more people but places them less precisely, raising MODA (95.4 -> 96.5) but dropping MODP (89.8 -> 75.0). Table 1 and Table 2 compare the detection and tracking performance of these methods.

Table 1. Comparison of the BEV detection performance on the Wildtrack dataset (results reported in [2]). †3DROM results are without additional data augmentations.

Method	Lifting method	MODA	MODP	Precision	Recall
DeepMCD	Learned	67.8	64.2	85	82
Deep-Occlusion	Learned	74.1	53.8	95	80
MVDet [4]	Perspective Transformation	88.2	75.7	94.7	93.6
SHOT	Perspective Transformation	90.2	76.5	96.1	94.0
3DROM†	Perspective Transformation	91.2	76.9	95.9	95.3
MVDeTr	Perspective Transformation	91.5	82.1	97.4	94.0
EarlyBird [1]	Perspective Transformation	91.2	81.8	94.9	96.3

Method	Lifting method	MODA	MODP	Precision	Recall
MVTT	Perspective Transformation + Region of Interest	94.1	81.3	97.6	96.5
TrackTacular [2]	Perspective Transformation	91.8	79.8	96.2	95.6
TrackTacular [2]	Bilinear Sampling	92.1	76.2	97.0	95.1
TrackTacular [2]	Depth Splatting	93.2	77.5	97.3	95.8
TrackTacular [2]	Deformable Attention	78.4	73.1	93.8	84.0

Table 2. Evaluation of pedestrian tracking results on the Wildtrack dataset (results reported in [2]). †Re-computed by the authors of [2]. The GLMB-YOLOv3 and GLMB-DO values are reproduced as reported in [2]; their MT and ML add up to more than 100%, so these values may contain an error in the source.

Method	IDF1 ↑	MOTA ↑	MOTP ↑	MT ↑	ML ↓
KSP-DO [3]	73.2	69.6	61.5	28.7	25.1
KSP-DO-ptrack [3]	78.4	72.2	60.3	42.1	14.6
GLMB-YOLOv3	74.3	69.7	73.2	79.5	21.6
GLMB-DO	72.5	70.1	63.1	93.6	22.8
DMCT	77.8	72.8	79.1	61.0	4.9
DMCT Stack	81.9	74.6	78.9	65.9	4.9
ReST†	86.7	84.9	84.1	87.8	4.9
EarlyBird [1]	92.3	89.5	86.6	78.0	4.9
MVFlow	93.5	91.3	–	–	–
TrackTacular [2] (Perspective Transform)	94.2	89.6	81.7	87.8	4.9
TrackTacular [2] (Bilinear Sampling)	95.3	91.8	85.4	87.8	4.9
TrackTacular [2] (Depth Splatting)	93.6	90.2	84.2	82.9	4.9
TrackTacular [2] (Deformable Attention)	88.0	82.2	78.9	75.6	4.9

4. Proposed Research Approach

Retrack builds on TrackTacular [2], which fuses seven overlapping camera views into one top-down Bird's Eye View (BEV) map and tracks pedestrians on it. Our reproduction on WILDTRACK [3] showed that most errors are missed people, so Retrack is a single hybrid pipeline that targets them at three stages. At detection, we tune the confidence threshold, which the baseline fixes at a default, to recover missed people at a small cost in false alarms. At association, following ByteTrack [5], low-confidence detections continue existing tracks but never start new ones, and inspired by EarlyBird [1], each track stores an appearance description from a pretrained re-identification model so that a person who reappears after being hidden keeps the same track ID. At view fusion, we compare the baseline, a version retrained with cameras randomly removed, and a version that averages the cameras seeing each location, each tested with one to three cameras removed. We expect fewer missed people, giving higher MODA and MOTA than our reproduced baseline, fewer broken tracks, and a more gradual loss of accuracy as cameras are removed.

5. System Requirements Specification

5.1 System Features

Retrack exposes a controlled workflow around the trained multi-view backend. The feature groups below map to the functional requirements in Section 5.2.

Table 12 System Features

Feature	Description	Requirements
Accounts and profiles	Registration, approval, authentication and approved profile edits.	FR-01 to FR-09
Configuration	Calibration validation, checkpoint selection and initialization.	FR-10 to FR-13
Monitoring	Input modes, ordered frames, detection, tracking, display, structured results and completion.	FR-14 to FR-24
Reports	Eligible automatic/manual generation, summaries and file retrieval.	FR-25 to FR-29
Audit	Diagnostics and administrative records.	FR-30

5.2 Functional Requirements

Protected operations require an authenticated, permitted account and authorized access to the affected resource. A required error response shall identify the failed condition without exposing secrets or internal server details.

5.2.1 Registration and Account Access

FR-01 Registration

The system shall validate an End User registration containing name, username, email and password, reject duplicate username or email values, and create a valid account as Pending. Public registration shall create an End User account only.

FR-02 Account access

Only an Active account shall obtain authenticated access. Pending and Rejected accounts shall be refused with an account-status message without exposing password information.

Acceptance: A valid new registration is Pending; a duplicate submission creates no account. Pending and Rejected credentials cannot establish access.

5.2.2 Authentication and Authorization

FR-03 Login

The system shall verify submitted credentials and account status before displaying the dashboard for the account role. A failed attempt shall not create an authenticated session.

FR-04 Protected access

Every protected request shall verify the authenticated session and the permission required for the operation. End Users shall access their own sessions and reports; Admin shall access administrative queues, configuration, logs and reports.

FR-05 Logout

Logout shall invalidate the current authentication session and return the browser to the login page. Logout shall not silently stop a running analysis session; stopping monitoring is a separate operation.

Acceptance: Successful login opens only the role dashboard. End Users cannot call Admin operations or access another user's reports. Logout invalidates access without silently stopping inference.

5.2.3 Account Approval

FR-06 Account approval

An authenticated Admin shall list Pending registrations and approve or reject a selected account. Approval shall change Pending to Active; rejection shall change Pending to Rejected. The system shall record the decision, Admin and time, and refuse a second decision against an already resolved registration.

Acceptance: An Admin decision changes a Pending account once; a second decision is refused and the first remains authoritative.

5.2.4 Profile Requests and Approved Updates

FR-07 Profile request

An End User shall submit a profile-update request identifying the fields to change and a reason, and shall be able to view its Pending, Approved, Rejected or Used status.

FR-08 Profile decision

An Admin shall approve or reject a pending profile-update request. The decision, permitted fields, deciding Admin and decision time shall be recorded.

FR-09 Approved profile editing

The system shall allow an End User to update only fields authorized by an unused Approved request. A successful save shall mark the request Used. Invalid values, duplicate account identifiers, unauthorized fields or reuse shall leave the profile unchanged and shall not consume an otherwise valid approval. Role and account status shall not be editable through this workflow.

Acceptance: Only unused Approved requests authorize the listed fields. Invalid saves leave the request available; successful saves mark it Used.

5.2.5 Calibration Input and Validation

FR-10 Calibration input

An End User shall upload or select a stored calibration bundle for the configured seven-camera rig before starting monitoring. The session shall retain the calibration reference and camera order used.

FR-11 Calibration validation

The system shall require one intrinsic and one extrinsic calibration for each configured camera, validate matrix and vector dimensions and finite numeric values, and require a consistent coordinate convention and translation unit. Invalid calibration shall prevent session start.

Acceptance: Missing or malformed calibration prevents Start and identifies the affected camera/file.

5.2.6 Model Checkpoint Configuration

FR-12 Checkpoint administration

An Admin shall select a trained checkpoint compatible with the camera count, scene configuration, backend and output contract. Unavailable or incompatible checkpoints shall be rejected. A checkpoint change shall be refused while its model instance is processing an active session.

Acceptance: An unavailable or incompatible checkpoint is refused. An active model instance cannot change a checkpoint.

5.2.7 Session Initialization and Input Modes

FR-13 Session initialization

Before accepting processing input, the system shall validate session configuration, load the selected checkpoint and reset tracking state for the new session. Failed initialization shall be recorded and shall not leave a session marked Running.

FR-14 Input modes

The system shall support LIVE input and RECORDED replay through MultiCameraFeed. Both modes shall provide synchronized frame sets from the same configured seven-camera rig; recorded input shall preserve source order and terminate at end of source.

Acceptance: Failed initialization never yields Running. LIVE and RECORDED fixtures both use the same input contract; replay ends explicitly.

5.2.8 Frame Validation and Ordered Processing

FR-15 Frame validation

Each accepted FrameSet shall contain exactly one decodable image for each configured camera, in the configured camera order, at 1920 by 1080 pixels, with a common declared frame index and source time. Duplicate, missing, wrongly sized or mismatched input shall not reach inference.

FR-16 Ordered processing

Accepted frame sets shall be processed in increasing source order. Rejected or skipped input shall not increment framesProcessed. The input adapter shall preserve synchronization metadata and shall not pair unrelated frames merely because they arrived together.

FR-17 Input faults

Invalid frame sets shall be skipped and recorded. A missing required camera shall pause inference until a complete valid set is available or Stop is requested. The operator shall see a diagnostic status and shall remain able to request Stop.

Acceptance: Missing, duplicate, wrongly sized, mismatched or out-of-order data never reaches inference or increases framesProcessed.

5.2.9 Pedestrian Detection and Temporal Tracking

FR-18 Pedestrian inference

The selected backend shall extract and aggregate calibrated image features into a common bird eye view, detect pedestrian ground positions, and return confidence values under its documented output convention. Other object or behavior categories are outside this prototype.

FR-19 Temporal tracking

The system shall associate detections across processed frame sets, create or update Track records and maintain trajectory samples. Track identifiers shall be unique within a session and shall not claim a real identity or continuity across separate sessions.

Acceptance: Ground-position detections and tracks follow the selected output convention. A new session resets tracking state.

5.2.10 Result Storage and Visualization

FR-20 Result persistence

For each successfully processed frame set, the system shall store sessionId, frameIndex, timestamp, track identifiers, ground positions, confidence and the frame occupancy needed to reproduce reports. A processed frame with no observed pedestrians shall still be stored with occupancy zero.

FR-21 Tracking display

An End User shall view updated camera annotations with track identifiers and a bird eye view showing ground positions and trajectories. The display shall identify its session and frame time, and shall work with either input mode.

FR-22 Structured result exchange

The backend shall provide a structured tracking-result response containing session, model, frameSet and results objects. It shall identify the session and input mode, model/checkpoint identifier, camera order, source frame index/time, observed track IDs, ground positions, confidence and occupancy. Responses shall declare coordinate units and shall not expose credentials, password hashes or internal server paths.

Acceptance: A valid zero-detection frame is committed with occupancy zero. Camera/BEV updates identify the session and source time.

5.2.11 Session Completion and Failure Handling

FR-23 Session completion

On an authorized Stop request or recorded-source exhaustion, the system shall stop accepting new input, release the feed and finalize the session. A Completed session with committed processed results shall invoke automatic report generation. An empty session shall return its summary without generating a report. Stop shall remain available while waiting for input.

FR-24 Processing failure

A fatal model or resource error shall terminate processing, mark the session Failed, release resources, preserve previously committed results and record a diagnostic visible to Admin. Finalization shall attempt automatic partial-report generation when committed processed results exist; the report shall clearly identify the Failed session and its partial coverage.

Acceptance: Stop and recorded EOF finalize normally; injected fatal failure preserves prior committed results and marks Failed.

5.2.12 Report Generation and Summary

FR-25 Report generation

The system shall attempt automatic report generation at finalization of a Completed or Failed session that has committed processed results. It shall also accept authorized manual generation requests for such sessions from the owning End User or Admin. Inaccessible, still-running or empty sessions shall be refused; a failed report-generation attempt shall leave the finalized session results available for a later retry.

FR-26 Report summary

Each report shall identify its session, final status, input mode, source coverage and generation time, and contain framesProcessed, uniqueTracks, meanOccupancy and peakOccupancy computed only from successfully processed, committed frame records. Valid frames with no observed pedestrians shall contribute occupancy zero. Rejected input and unobserved predicted tracks shall not increase occupancy. uniqueTracks shall be labelled as session-local track IDs, not guaranteed unique real people.

FR-27 Report formats

The system shall create PDF and CSV report outputs with consistent summary figures. CSV shall include summary and frame-level data sufficient to reproduce occupancy figures; schema and units shall be documented. Rendering failures shall be recorded and shall not produce a falsely available report.

Acceptance: Only eligible finalized sessions generate reports. PDF and CSV summary values agree; empty sessions are refused.

5.2.13 Report Retrieval and Audit Records

FR-28 Report retrieval

End User and Admin shall search or view authorized reports using session identifiers and date ranges. An invalid date range shall be rejected; a valid query with no matches shall return an empty-result state.

FR-29 Report download

End User and Admin shall download an authorized available report in PDF or CSV format. Access shall be checked before delivering the file; an unavailable file shall return a clear error. Successful downloads shall be recorded in the audit log.

FR-30 Audit and diagnostics

The system shall record account and profile decisions, checkpoint changes, session start and completion, report generation and downloads, and processing or input errors with time, actor or system identity, affected resource and outcome. Admin shall be able to inspect these records.

Acceptance: Search respects access/date boundaries; available authorized downloads are audited. Diagnostics contain no credentials.

5.3 Non Functional Requirements

The following requirements define quality and operating expectations. The performance figures are proposed review targets, not measured deployment results. Tests must record hardware, model/checkpoint configuration, browser versions, source rate and load.

5.3.1 User Friendly Interface

NFR-01 Usability

The application shall provide readable session, account, profile-request and report states on a desktop display. Validation errors shall identify the affected camera, field or resource and an actionable reason.

The interface shall distinguish waiting for input, invalid setup, processing failure, unavailable reports and valid processing with zero pedestrians. Viewing a result or report shall be optional and shall not gate processing.

5.3.2 Data Security

NFR-02 Credential protection

Passwords shall be stored as salted one-way hashes. Passwords, authentication secrets and raw session tokens shall not be written to logs or result payloads.

NFR-03 Permission enforcement

Role and resource authorization shall be enforced on the server. Changing a browser request or hiding a UI control shall not bypass account approval, profile approval or report ownership checks.

NFR-04 Transport protection

Deployment outside a local development environment shall protect credentials and authenticated traffic using HTTPS. Client-facing errors shall not expose server paths or stack traces.

5.3.3 Performance

NFR-05 Command responsiveness

The proposed prototype target is a 95th-percentile acknowledgement time of at most 2 seconds for ordinary session commands under the declared single-session load, excluding model loading and report rendering. Results shall be measured, not assumed.

NFR-06 Processing performance

The proposed benchmark target is 2 synchronized frame sets per second with at most 2 seconds of 95th-percentile result delay after warm-up, on the documented CUDA reference host and selected checkpoint. If unmet, measured throughput and the supported replay rate shall be reported; real-time capability shall not be claimed.

NFR-07 Initial capacity

The initial deployment shall run one active analysis session per model instance. A second start against a busy instance shall be rejected clearly; historical session and report access shall remain available.

5.3.4 Privacy

NFR-08 Data minimization and retention

The application shall restrict pedestrian tracking results and exports to authorized users, avoid face recognition and real-person identity fields, and document the deployment retention/deletion policy. Raw images shall not be retained automatically solely to generate session reports. Stored positions and session-local IDs remain sensitive movement data and shall not be described as inherently anonymous.

5.3.5 Compatibility

NFR-09 Browser compatibility

Account, monitoring and reporting workflows shall be verified on the selected current Chromium-based and Firefox desktop browsers, with the tested versions recorded in the test evidence.

The selected inference environment, repository revision and checkpoint shall be pinned. Initial application acceptance shall target the specified seven-camera scene; wider scene or camera support requires a separate compatibility evaluation.

5.3.6 Reliability and Persistent Storage

NFR-10 Persistence

Committed session records and generated reports shall use configured persistent storage outside an ephemeral inference runtime. Finalization and report publication shall not expose partially written records as complete.

NFR-11 Reliability

A processing error shall not reuse another session tracking state or erase prior committed results. Zero detections shall be treated as valid inference output, while absent input shall be treated as an input fault.

5.3.7 Reproducibility

NFR-12 Inference reproducibility

Evaluation shall record the repository revision, model configuration, checkpoint identity, dataset split, preprocessing, source order, environment and hardware. Web-wrapper replay and offline comparison shall use the same inputs and thresholds.

NFR-13 Evaluation honesty

Acceptance evidence shall distinguish offline model evaluation from web-system tests. No production availability or accuracy guarantee shall be inferred from a research-paper result.

5.3.8 Maintainability

NFR-14 Maintainability

Web presentation, persistence and model integration shall be separated. AIModel shall expose loadCheckpoint, resetTracking and analyze operations without requiring browser code to depend on research-repository internals.

5.3.9 Time and Units

NFR-15 Time and units

Result timestamps shall use a documented ISO 8601 convention. Ground positions shall use a declared shared ground-plane coordinate system and centimeters. Report filters shall interpret date boundaries consistently.

5.3.10 Backup and Recovery

NFR-16 Recovery and records

Reports and session metadata shall have a documented backup and recovery procedure. Recovery shall restore completed records; automatic resumption of interrupted GPU inference is outside the initial prototype.

6. Use Case Diagram



Figure 1 Use Case Diagram of the Retrack System

7. High Level Use Cases

7.1 Register Account

Table 13 High Level Use Case – Register Account

Use Case	UC-01 Register Account
Actors	End User (primary)
Type	Primary
Description	A new user submits registration details. The system creates an End User account in Pending status, and the account can be used only after an Admin approves it (UC-06).

7.2 Log In / Log Out

Table 14 High Level Use Case – Log In / Log Out

Use Case	UC-02 Log In / Log Out
Actors	End User, Admin
Type	Primary (includes UC-03 Authenticate User)
Description	A registered user logs in to reach the dashboard for their role and logs out to end the browser authentication session.

7.3 Authenticate User

Table 15 High Level Use Case – Authenticate User

Use Case	UC-03 Authenticate User
Actors	System (included by UC-02 and by every protected request)
Type	Included
Description	The system verifies credentials and account status at login, and verifies the authentication session and the required permission on every protected request.

7.4 Request Profile Update

Table 16 High Level Use Case – Request Profile Update

Use Case	UC-04 Request Profile Update
Actors	End User (primary); Admin (secondary, decides the request in UC-07)
Type	Primary (includes UC-05 Update Profile)

Description	An End User asks the Admin for permission to change specific profile fields, gives a reason and can follow the status of the request.

7.5 Update Profile

Table 17 High Level Use Case – Update Profile

Use Case	UC-05 Update Profile
Actors	End User
Type	Included (by UC-04)
Description	After a profile-update request is approved, the End User edits only the fields that the approval covers.

7.6 Approve / Reject Accounts

Table 18 High Level Use Case – Approve / Reject Accounts

Use Case	UC-06 Approve / Reject Accounts
Actors	Admin
Type	Primary
Description	The Admin reviews Pending registrations and approves or rejects each one.

7.7 Approve / Reject Profile Requests

Table 19 High Level Use Case – Approve / Reject Profile Requests

Use Case	UC-07 Approve / Reject Profile Requests
Actors	Admin
Type	Primary
Description	The Admin reviews Pending profile-update requests and approves or rejects each one.

7.8 Upload / Select Calibration

Table 20 High Level Use Case – Upload / Select Calibration

Use Case	UC-08 Upload / Select Calibration
Actors	End User
Type	Primary (includes UC-09 Validate Calibration)

Description	Before starting monitoring, the End User uploads a calibration bundle for the seven-camera rig or selects a stored one.

7.9 Validate Calibration

Table 21 High Level Use Case – Validate Calibration

Use Case	UC-09 Validate Calibration
Actors	System (included by UC-08)
Type	Included
Description	The system checks that a calibration bundle is complete and well formed before a session can use it.

7.10 Manage Model Checkpoint

Table 22 High Level Use Case – Manage Model Checkpoint

Use Case	UC-10 Manage Model Checkpoint
Actors	Admin
Type	Primary
Description	The Admin selects the trained checkpoint used for analysis; the system accepts only compatible checkpoints.

7.11 Start / Stop Monitoring Session

Table 23 High Level Use Case – Start / Stop Monitoring Session

Use Case	UC-11 Start / Stop Monitoring Session
Actors	End User
Type	Primary
Description	The End User starts a LIVE or RECORDED monitoring session with a validated calibration and later stops it; the system finalizes the session.

7.12 Supply Synchronized Frames

Table 24 High Level Use Case – Supply Synchronized Frames

Use Case	UC-12 Supply Synchronized Frames
Actors	Camera Rig (primary)
Type	Primary

Description	During a Running session the Camera Rig, or a recorded replay, supplies synchronized frame sets of seven images; the system validates and processes each frame set.

7.13 View Live Tracking Results

Table 25 High Level Use Case – View Live Tracking Results

Use Case	UC-13 View Live Tracking Results
Actors	End User
Type	Primary
Description	The End User watches the camera views annotated with track IDs and a bird’s-eye view showing ground positions and trajectories.

7.14 Generate Session Report

Table 26 High Level Use Case – Generate Session Report

Use Case	UC-14 Generate Session Report
Actors	End User, Admin; System (automatic at session finalization)
Type	Primary
Description	The system produces a PDF and CSV report for a finalized session from its committed frame records.

7.15 Search Reports

Table 27 High Level Use Case – Search Reports

Use Case	UC-15 Search Reports
Actors	End User, Admin
Type	Primary
Description	A user searches the stored reports they are allowed to see by session identifier and date range.

7.16 Download Report

Table 28 High Level Use Case – Download Report

Use Case	UC-16 Download Report
Actors	End User, Admin

Type	Primary
Description	A user downloads an available report as a PDF or CSV file.

7.17 View Audit Log and System Errors

Table 29 High Level Use Case – View Audit Log and System Errors

Use Case	UC-17 View Audit Log and System Errors
Actors	Admin
Type	Primary
Description	The Admin inspects the audit and diagnostic records of the system.

8. Expanded Use Cases

8.1 Register Account

Table 30 Expanded Use Case – Register Account

Use Case	UC-01 Register Account	
Actors	End User (primary)	
Description	A new user submits registration details. The system creates an End User account in Pending status, and the account can be used only after an Admin approves it (UC-06).	
Related Requirements	FR-01, FR-02, FR-06, NFR-02	
Preconditions	The person does not already have an account. The registration page is reachable.	
Postconditions	A Pending End User account is stored with a salted password hash.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User opens the registration page.	Displays the registration form: name, username, email and password.
Step 2	2. User fills in the form and submits it.	Validates the fields and checks that the username and email are not already in use.
Step 3		Stores the password as a salted one-way hash, creates the End User account as Pending and shows that the account is waiting for Admin approval.
Alternate Courses	2a. A field is missing or invalid: the system identifies the field and does not create an account. 2b. The username or email already exists: the system rejects the registration and no account is created.	
Constraints	Public registration creates an End User account only. Admin accounts are provisioned separately and cannot be created or promoted through registration.	

8.2 Log In / Log Out

Table 31 Expanded Use Case – Log In / Log Out

Use Case	UC-02 Log In / Log Out
Actors	End User, Admin
Description	A registered user logs in to reach the dashboard for their role and logs out to end the browser authentication session.

Related Requirements	FR-02, FR-03, FR-05	
Preconditions	Log in: the user has an account. Log out: the user is logged in.	
Postconditions	Log in: an authenticated session exists and the role dashboard is shown. Log out: the authentication session is invalidated and the login page is shown.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User opens the login page, enters their credentials and submits them.	Passes the credentials to Authenticate User (UC-03).
Step 2		On success, creates the authentication session and opens the dashboard for the user's role.
Step 3	3. User selects Log Out.	Invalidates the current authentication session.
Step 4		Returns the browser to the login page.
Alternate Courses	2a. The credentials are invalid: the system shows an error and does not create an authenticated session. 2b. The account is Pending or Rejected: access is refused with an account-status message that does not expose password information. 3a. The authentication session has already expired: the system returns the browser to the login page.	
Constraints	Logging out does not stop a running analysis session. Stopping monitoring is a separate operation (UC-11).	

8.3 Authenticate User

Table 32 Expanded Use Case – Authenticate User

Use Case	UC-03 Authenticate User	
Actors	System (included by UC-02 and by every protected request)	
Description	The system verifies credentials and account status at login, and verifies the authentication session and the required permission on every protected request.	
Related Requirements	FR-02, FR-03, FR-04, NFR-02, NFR-03	
Preconditions	A login attempt or a protected request has been received.	
Postconditions	Access is granted for the account's role, or refused.	
Typical Course of Events	**Actor Action**	**System Response**

Step 1	1. User submits credentials (login) or a protected request.	Looks up the account and verifies the password against the stored salted hash, or validates the authentication session.
Step 2		Checks that the account status is Active.
Step 3		Checks that the role and resource permissions allow the requested operation.
Step 4		Grants access and returns the account identity and role to the caller.
Alternate Courses	1a. The account is unknown or the password is wrong: the system returns an invalid-credentials error without saying which part was wrong. 1b. The authentication session is invalid or expired: the request is refused and the browser is returned to the login page. 2a. The account is Pending or Rejected: access is refused. 3a. The operation is not permitted, for example an End User calling an Admin operation or opening another user's report: the request is refused.	
Constraints	Authorization is enforced on the server; changing a browser request or hiding a UI control cannot bypass it. Passwords, secrets and raw session tokens are never written to logs.	

8.4 Request Profile Update

Table 33 Expanded Use Case – Request Profile Update

Use Case	UC-04 Request Profile Update	
Actors	End User (primary); Admin (secondary, decides the request in UC-07)	
Description	An End User asks the Admin for permission to change specific profile fields, gives a reason and can follow the status of the request.	
Related Requirements	FR-07, FR-08	
Preconditions	The End User is logged in.	
Postconditions	A Pending profile-update request is stored with the requested fields and the reason.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User opens Request Profile Update.	Displays the profile fields that can be requested and a reason box.
Step 2	2. User selects the fields, enters a reason and submits the request.	Validates the request, stores it as Pending and confirms that it was submitted.

Step 3	3. User later opens the request.	Shows its status: Pending, Approved, Rejected or Used.
Alternate Courses	2a. No field is selected or the reason is missing: the system asks the user to complete the request and stores nothing. 3a. The Admin approved the request: the user can now edit the approved fields (UC-05).	
Constraints	Role and account status cannot be requested or changed through this workflow.	

8.5 Update Profile

Table 34 Expanded Use Case – Update Profile

Use Case	UC-05 Update Profile	
Actors	End User	
Description	After a profile-update request is approved, the End User edits only the fields that the approval covers.	
Related Requirements	FR-09	
Preconditions	The End User is logged in and has an Approved, unused profile-update request.	
Postconditions	The approved fields are updated and the request is marked Used.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User opens their profile.	Displays the profile with only the approved fields editable.
Step 2	2. User edits the permitted fields and saves.	Validates the new values.
Step 3		Stores the changes, marks the request Used and confirms the update.
Alternate Courses	1a. There is no Approved, unused request: the profile is read-only and the user is directed to Request Profile Update (UC-04). 2a. A value is invalid or the username or email is already in use: the profile is left unchanged, the error is shown and the approval remains available. 2b. A field outside the approval is changed: the save is refused and the profile is left unchanged.	
Constraints	Each approval can be used once. Role and account status are not editable through this workflow.	

8.6 Approve / Reject Accounts

Table 35 Expanded Use Case – Approve / Reject Accounts

Use Case	UC-06 Approve / Reject Accounts	
Actors	Admin	
Description	The Admin reviews Pending registrations and approves or rejects each one.	
Related Requirements	FR-06, FR-30	
Preconditions	The Admin is logged in and at least one Pending registration exists.	
Postconditions	The account is Active or Rejected, and the decision, deciding Admin and time are recorded.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. Admin opens the list of Pending registrations.	Lists the Pending accounts with their registration details.
Step 2	2. Admin selects an account and approves it.	Changes the status from Pending to Active.
Step 3		Records the decision, the deciding Admin and the time in the audit log.
Alternate Courses	2a. The Admin rejects the account: the status changes from Pending to Rejected and the decision is recorded. 2b. The account has already been decided: the second decision is refused and the first remains in effect.	
Constraints	Approval activates an End User account only; it cannot create or promote an Admin.	

8.7 Approve / Reject Profile Requests

Table 36 Expanded Use Case – Approve / Reject Profile Requests

Use Case	UC-07 Approve / Reject Profile Requests
Actors	Admin
Description	The Admin reviews Pending profile-update requests and approves or rejects each one.
Related Requirements	FR-08, FR-30
Preconditions	The Admin is logged in and at least one Pending profile-update request exists.
Postconditions	The request is Approved or Rejected, and the decision, permitted fields, deciding Admin and time are recorded.

Typical Course of Events	Actor Action	System Response
Step 1	1. Admin opens the profile-request queue.	Lists the Pending requests.
Step 2	2. Admin selects a request.	Shows the requester, the requested fields, the reason and the current values.
Step 3	3. Admin approves the request.	Sets the request to Approved with the permitted fields and records the decision, Admin and time.
Alternate Courses	3a. The Admin rejects the request: the status becomes Rejected, the decision is recorded and the End User sees the Rejected status.	
Constraints	An approval authorizes one update of the listed fields; it becomes Used after a successful save (UC-05).	

8.8 Upload / Select Calibration

Table 37 Expanded Use Case – Upload / Select Calibration

Use Case	UC-08 Upload / Select Calibration	
Actors	End User	
Description	Before starting monitoring, the End User uploads a calibration bundle for the seven-camera rig or selects a stored one.	
Related Requirements	FR-10, FR-11	
Preconditions	The End User is logged in.	
Postconditions	The session configuration references a validated calibration bundle and the camera order used.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User opens session setup.	Offers to upload a calibration bundle or select a stored one.
Step 2	2. User uploads a bundle or selects a stored bundle.	Runs Validate Calibration (UC-09) on the bundle.
Step 3		Attaches the calibration reference and camera order to the session configuration and shows that the calibration is valid.

Alternate Courses	3a. The calibration is invalid: the system shows the affected camera or file and the reason, Start remains unavailable, and the user corrects the setup and repeats step 2.
Constraints	Exactly seven fixed cameras in the order CVLab1 to CVLab4, then IDIAP1 to IDIAP3, with centimetre translation units. Uploading another calibration alone does not establish support for a new site.

8.9 Validate Calibration

Table 38 Expanded Use Case – Validate Calibration

Use Case	UC-09 Validate Calibration	
Actors	System (included by UC-08)	
Description	The system checks that a calibration bundle is complete and well formed before a session can use it.	
Related Requirements	FR-11, NFR-01	
Preconditions	A calibration bundle has been uploaded or selected.	
Postconditions	The bundle is marked valid, or rejected with a list of problems.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1		Checks that there is one intrinsic and one extrinsic calibration for each of the seven configured cameras.
Step 2		Checks matrix and vector dimensions and that all values are finite numbers.
Step 3		Checks for one shared ground-plane coordinate convention and centimetre translation units.
Step 4		Marks the bundle valid and returns it to UC-08.
Alternate Courses	1a. A calibration file is missing: the system reports the camera and the missing file. 2a. A matrix or vector is malformed or contains a non-finite value: the system reports the camera and the field. 3a. The convention or unit is inconsistent: the system reports the problem. In every case the session cannot start.	
Constraints	Validation checks structure and values; it cannot prove that the calibration matches the physical scene. Moving a camera invalidates its calibration.	

8.10 Manage Model Checkpoint

Table 39 Expanded Use Case – Manage Model Checkpoint

Use Case	UC-10 Manage Model Checkpoint	
Actors	Admin	
Description	The Admin selects the trained checkpoint used for analysis; the system accepts only compatible checkpoints.	
Related Requirements	FR-12, FR-30	
Preconditions	The Admin is logged in and trained checkpoints are available.	
Postconditions	The selected checkpoint is active for new sessions and the change is recorded.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. Admin opens checkpoint management.	Lists the available checkpoints and shows the active one.
Step 2	2. Admin selects a checkpoint.	Checks compatibility with the camera count, scene configuration, backend and output contract.
Step 3		Makes the checkpoint active and records the change, the Admin and the time.
Alternate Courses	2a. The checkpoint is unavailable or incompatible: the change is refused with the reason and the previous checkpoint stays active. 2b. The model instance is processing an active session: the change is refused until that session ends.	
Constraints	A checkpoint is specific to its trained scene. A different scene or camera layout needs new calibration and a compatible checkpoint.	

8.11 Start / Stop Monitoring Session

Table 40 Expanded Use Case – Start / Stop Monitoring Session

Use Case	UC-11 Start / Stop Monitoring Session
Actors	End User
Description	The End User starts a LIVE or RECORDED monitoring session with a validated calibration and later stops it; the system finalizes the session.
Related Requirements	FR-13, FR-14, FR-23, FR-24, NFR-07

Preconditions	The End User is logged in, a validated calibration is selected (UC-08), a compatible checkpoint is active and the model instance has no other active session.	
Postconditions	Start: the session is Running with a fresh tracking state. Stop: the session is finalized and a report is generated automatically when committed results exist.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User chooses the input mode (LIVE or RECORDED) and selects Start.	Validates the session configuration, loads the selected checkpoint and resets the tracking state.
Step 2		Marks the session Running, starts accepting frame sets (UC-12) and opens the live view (UC-13).
Step 3	3. User selects Stop.	Stops accepting new input, releases the feed and finalizes the session.
Step 4		If committed results exist, runs Generate Session Report (UC-14) automatically and shows the session summary.
Alternate Courses	1a. Initialization fails: the failure is recorded, the session is not marked Running and the error is shown. 1b. The model instance is busy with another session: Start is refused clearly; reports remain available. 3a. A RECORDED source reaches its end: the session finalizes as if Stop had been selected. 3b. A fatal model or resource error occurs: the session is marked Failed, earlier results are kept, a diagnostic is recorded for the Admin and a partial report is attempted. 4a. No frame set was processed: the summary is shown without generating a report.	
Constraints	One active analysis session per model instance. Stop remains available while the session is waiting for input.	

8.12 Supply Synchronized Frames

Table 41 Expanded Use Case – Supply Synchronized Frames

Use Case	UC-12 Supply Synchronized Frames
Actors	Camera Rig (primary)
Description	During a Running session the Camera Rig, or a recorded replay, supplies synchronized frame sets of seven images; the system validates and processes each frame set.

Related Requirements	FR-15 to FR-22	
Preconditions	A session is Running with a validated calibration.	
Postconditions	Each valid frame set is processed in order and its results are stored; invalid frame sets are skipped and recorded.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. Camera Rig sends a frame set: seven images with a frame index and source time.	Checks for one decodable image per camera, the configured camera order, 1920 by 1080 pixels and a common frame index and source time.
Step 2		Accepts the frame set and processes frame sets in increasing source order.
Step 3		Extracts image features from all camera views, fuses them into the bird’s-eye view, detects pedestrians on the ground plane and associates detections with session-local tracks.
Step 4		Stores the ground positions, track IDs, confidence values and occupancy, and publishes the camera and bird’s-eye view results to the live view.
Step 5	5. Camera Rig sends the next frame set.	Repeats steps 1 to 4 until the session stops.
Alternate Courses	1a. The frame set is incomplete, duplicated, wrongly sized or mismatched: it is rejected with a reason, recorded and not counted in framesProcessed; the session continues. 1b. A required camera is missing: inference pauses, a diagnostic status is shown, and processing resumes when a complete valid frame set arrives or the user selects Stop. 3a. A valid frame set contains no pedestrians: the result is stored with occupancy zero.	
Constraints	Exactly seven fixed, synchronized cameras at 1920 by 1080 pixels. Pedestrians only. Track IDs are unique within one session and are not real identities.	

8.13 View Live Tracking Results

Table 42 Expanded Use Case – View Live Tracking Results

Use Case	UC-13 View Live Tracking Results
Actors	End User

Description	The End User watches the camera views annotated with track IDs and a bird’s-eye view showing ground positions and trajectories.	
Related Requirements	FR-21, FR-22, NFR-01	
Preconditions	The End User is logged in and owns a Running session.	
Postconditions	No data changes; viewing does not affect processing.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User opens the live view for their session.	Shows the camera views with track IDs and the bird’s-eye view with ground positions and trajectories, labelled with the session and frame time.
Step 2	2. User keeps watching.	Updates the views as each frame set is processed.
Alternate Courses	1a. The session is waiting for input: the view shows a waiting status. 1b. A valid frame set contains no pedestrians: the view shows zero pedestrians rather than an error. 1c. There is an input fault or processing failure: the view shows the corresponding status. 1d. The user opens another user’s session: access is refused.	
Constraints	Viewing results is optional and never holds up processing.	

8.14 Generate Session Report

Table 43 Expanded Use Case – Generate Session Report

Use Case	UC-14 Generate Session Report	
Actors	End User, Admin; System (automatic at session finalization)	
Description	The system produces a PDF and CSV report for a finalized session from its committed frame records.	
Related Requirements	FR-25, FR-26, FR-27	
Preconditions	The session is Completed or Failed and has committed processed results. A manual request comes from the owning End User or an Admin.	
Postconditions	PDF and CSV reports are stored and available for search and download.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. The session is finalized, or the user requests a report.	Checks that the session is eligible and that the requester may access it.

Step 2		Computes framesProcessed, uniqueTracks, meanOccupancy and peakOccupancy from the committed frame records.
Step 3		Creates the PDF and CSV reports with the same summary figures, identifying the session, final status, input mode, source coverage and generation time.
Step 4		Stores the reports and marks them available.
Alternate Courses	1a. The session is still running, empty or not accessible to the requester: the request is refused with the reason. 2a. The session Failed: the report clearly identifies the Failed status and its partial coverage. 3a. Rendering fails: the failure is recorded, the report is not marked available and the session results stay available for a later retry.	
Constraints	uniqueTracks counts session-local track IDs, not guaranteed unique real people. Rejected input does not increase occupancy.	

8.15 Search Reports

Table 44 Expanded Use Case – Search Reports

Use Case	UC-15 Search Reports	
Actors	End User, Admin	
Description	A user searches the stored reports they are allowed to see by session identifier and date range.	
Related Requirements	FR-28	
Preconditions	The user is logged in.	
Postconditions	A list of matching reports the user may access is displayed.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User opens Reports and enters a session identifier, a date range or both.	Validates the filters.
Step 2	2. User submits the search.	Searches only the reports the user is authorized to see (an End User’s own sessions; authorized reports for an Admin) and lists the matches.

Step 3	3. User opens a result.	Shows the report summary and the download options.
Alternate Courses	1a. The date range is invalid, for example the start is after the end: the search is rejected. 2a. No report matches: the system shows an empty-result state.	
Constraints	Date boundaries are interpreted consistently with the documented ISO 8601 time convention.	

8.16 Download Report

Table 45 Expanded Use Case – Download Report

Use Case	UC-16 Download Report	
Actors	End User, Admin	
Description	A user downloads an available report as a PDF or CSV file.	
Related Requirements	FR-27, FR-29, FR-30	
Preconditions	The user is logged in and an available report exists that they may access.	
Postconditions	The file is delivered and the download is recorded in the audit log.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. User selects a report and chooses PDF or CSV.	Checks that the user may access the report.
Step 2		Loads the file in the chosen format and returns it.
Step 3		Records the download in the audit log.
Alternate Courses	1a. The user may not access the report: the download is refused. 2a. The file is unavailable: the system returns a clear error.	
Constraints	The PDF and CSV files carry the same summary figures.	

8.17 View Audit Log and System Errors

Table 46 Expanded Use Case – View Audit Log and System Errors

Use Case	UC-17 View Audit Log and System Errors
Actors	Admin
Description	The Admin inspects the audit and diagnostic records of the system.

Related Requirements	FR-17, FR-24, FR-30	
Preconditions	The Admin is logged in.	
Postconditions	No data changes.	
Typical Course of Events	**Actor Action**	**System Response**
Step 1	1. Admin opens the audit log.	Shows records of account and profile decisions, checkpoint changes, session start and completion, report generation and downloads, and processing or input errors, each with time, actor or system identity, affected resource and outcome.
Step 2	2. Admin filters the records by type or time.	Shows the matching records.
Step 3	3. Admin opens an error record.	Shows the diagnostic for the failed session or input fault.
Alternate Courses	2a. No record matches: the system shows an empty-result state.	
Constraints	Only the Admin can view these records. Records never contain passwords, authentication secrets or session tokens.	

9. Activity Diagram

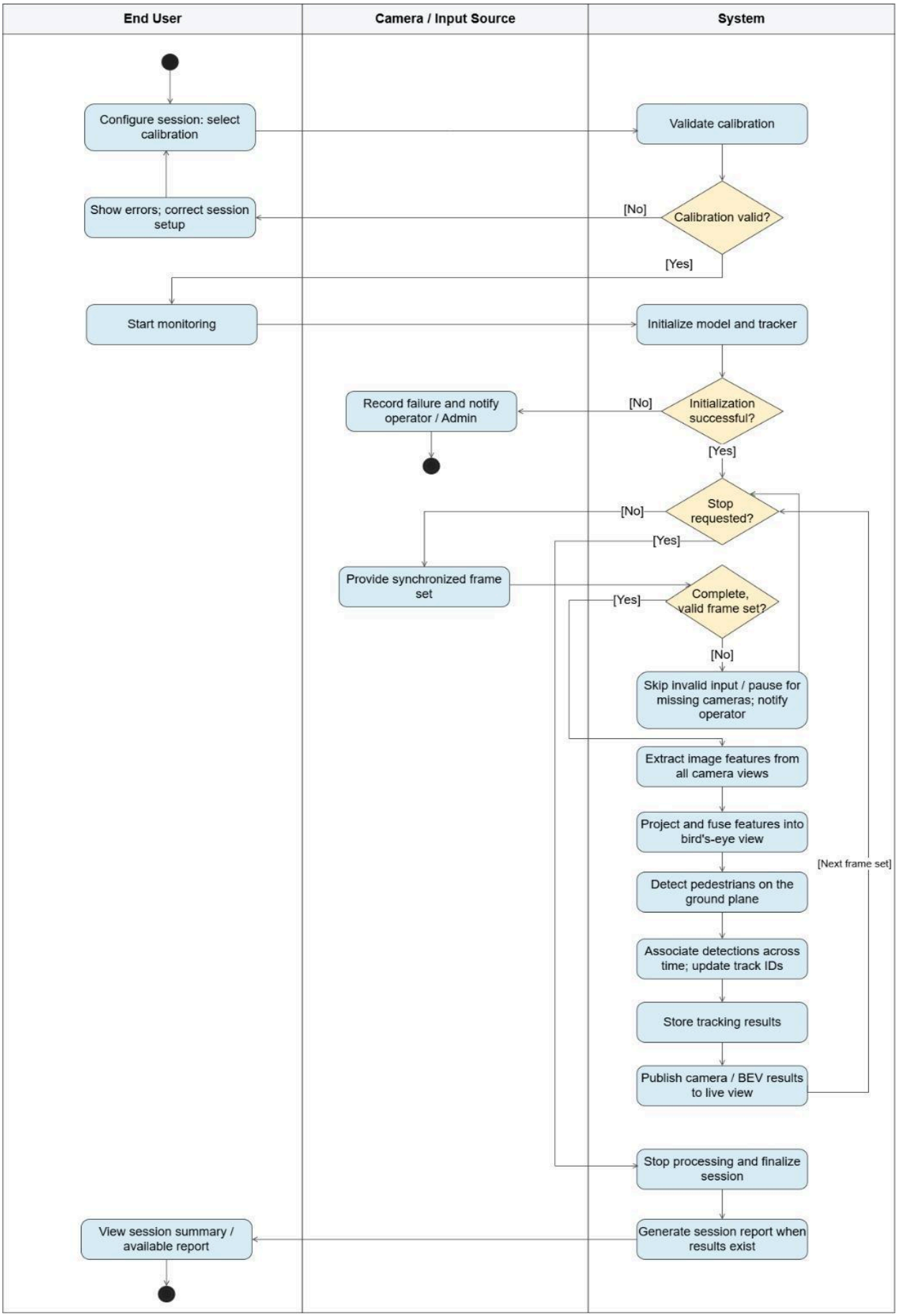


Figure 2 Activity Diagram of the Retrack Monitoring Workflow

10. System Sequence Diagram

10.1 Create User Account

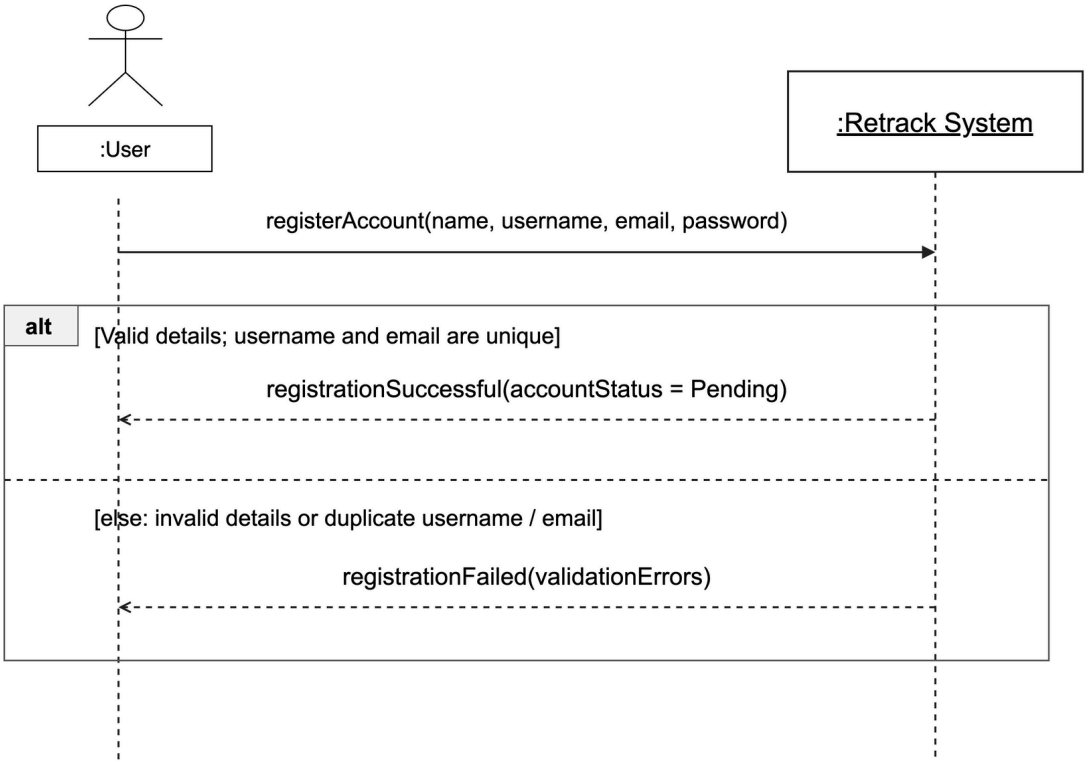


Figure 3 System Sequence Diagram – Create User Account

10.2 Request Profile Update

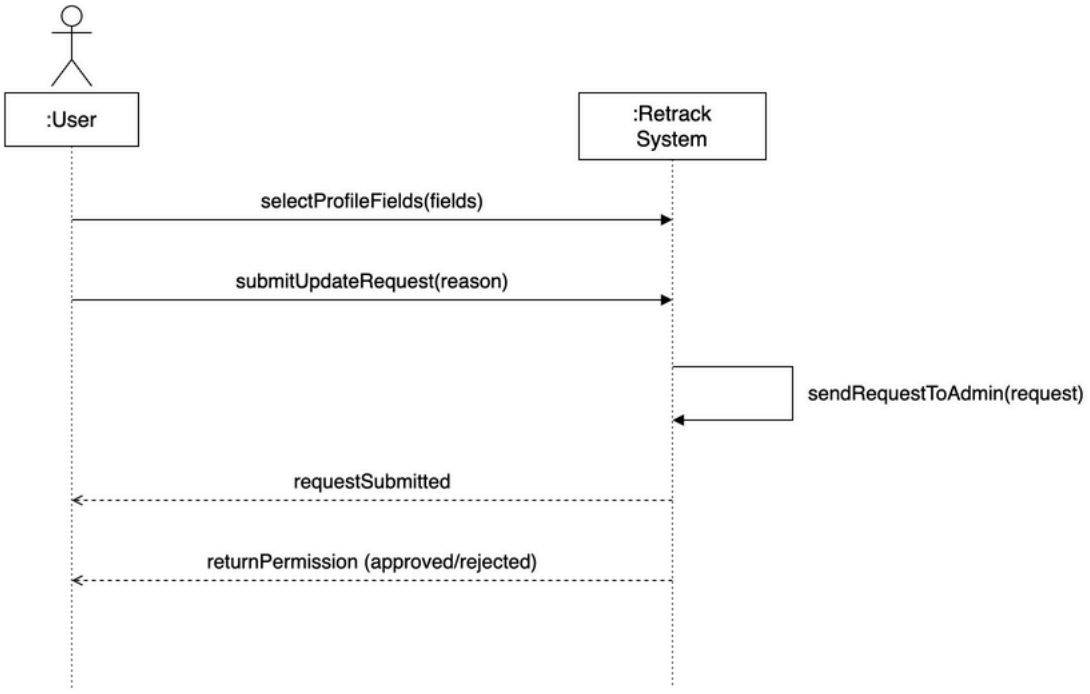


Figure 4 System Sequence Diagram - Request Profile Update

10.3 Update Profile

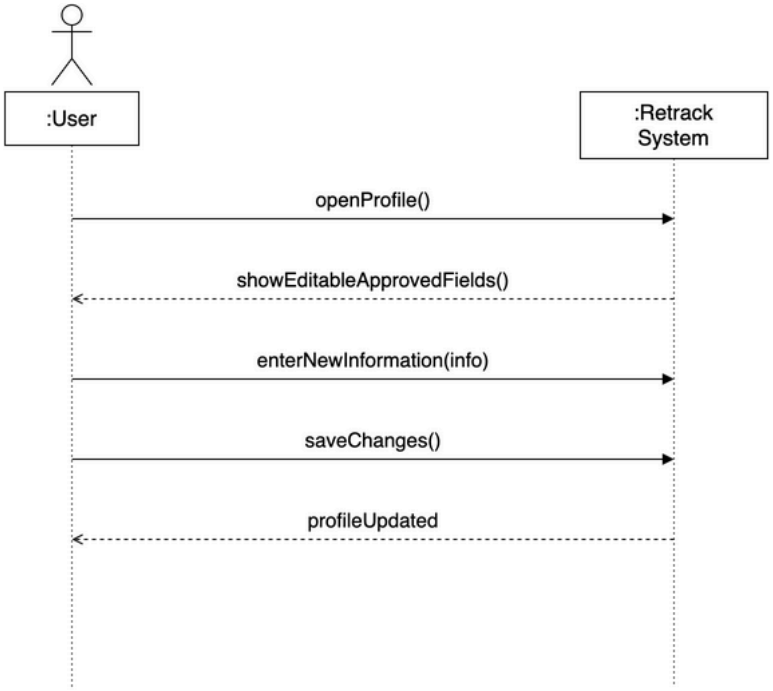


Figure 5 System Sequence Diagram – Update Profile

10.4 Search Report

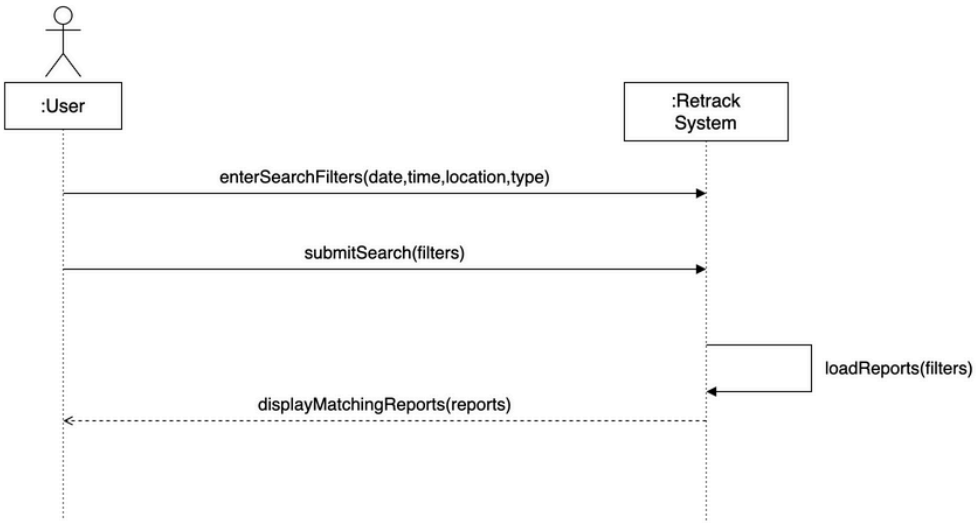


Figure 6 System Sequence Diagram – Search Report

10.5 Download Report

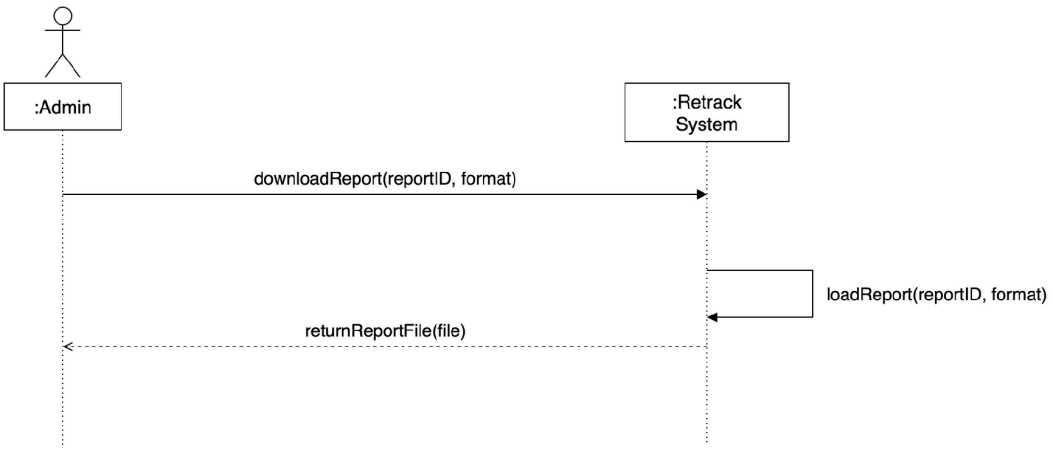


Figure 7 System Sequence Diagram – Download Report

10.6 Input live feed

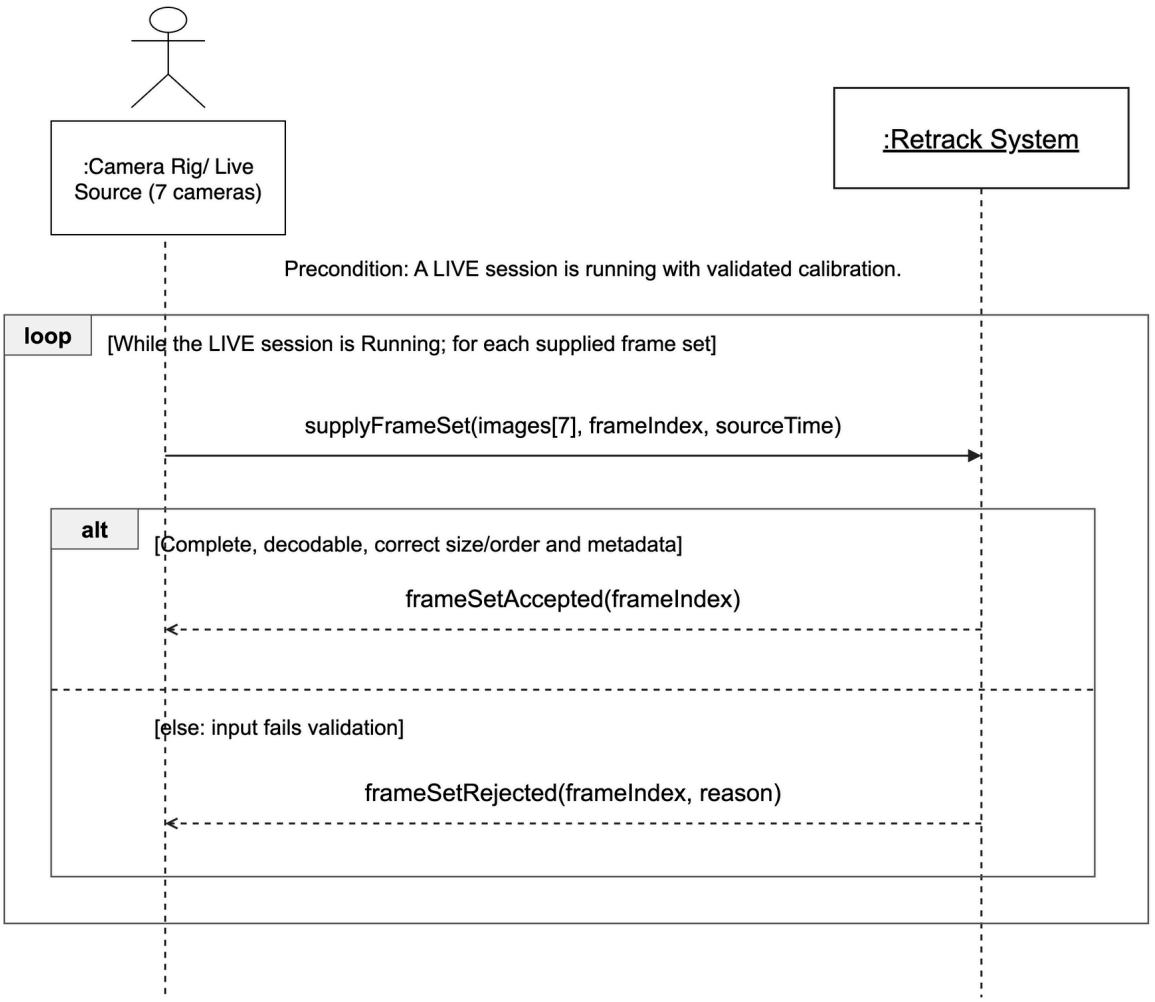


Figure 8 System Sequence Diagram – Input Live Feed

10.7 Logout

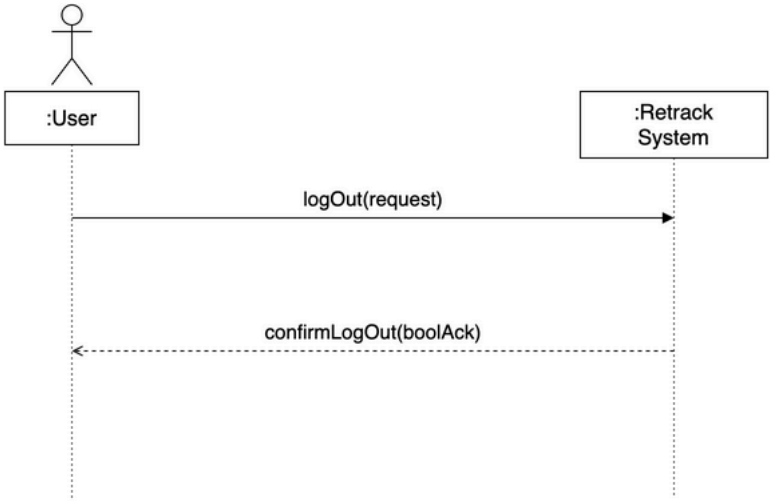


Figure 9 System Sequence Diagram – Logout

11. Domain Model

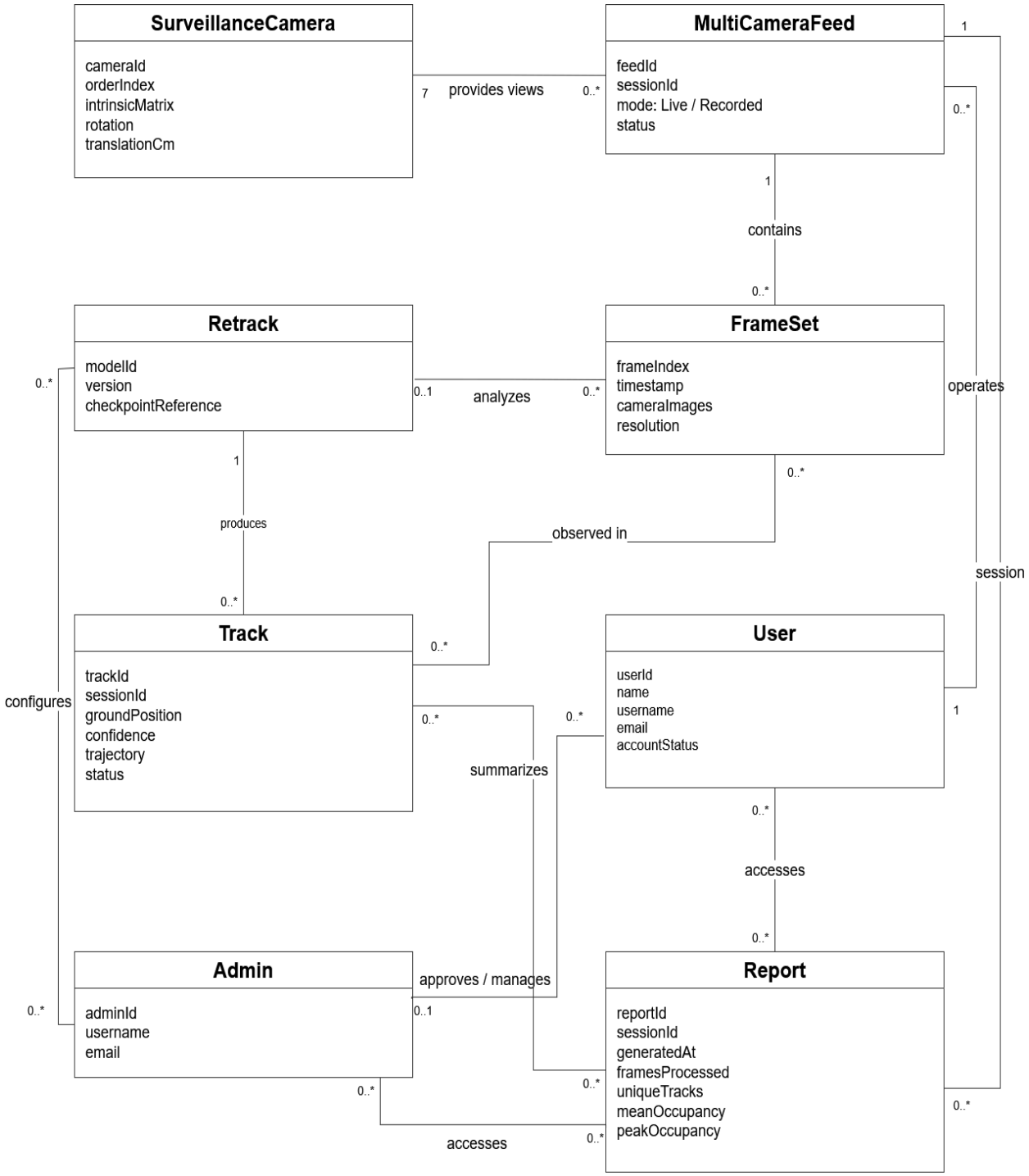


Figure 10 Domain Model of the Retrack System

12. Class Diagram

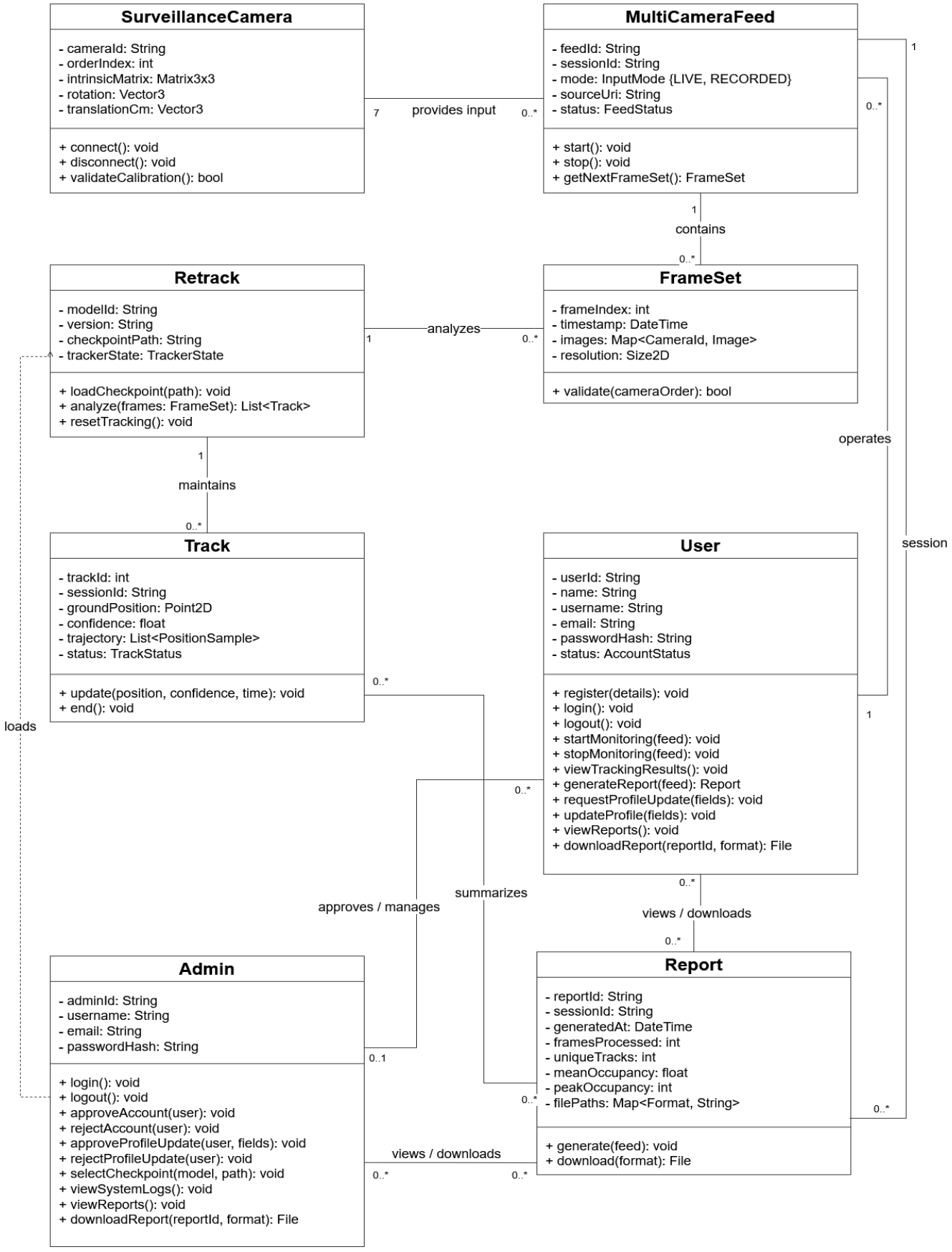


Figure 11 Class Diagram of the Retrack System

13. Sequence Diagram

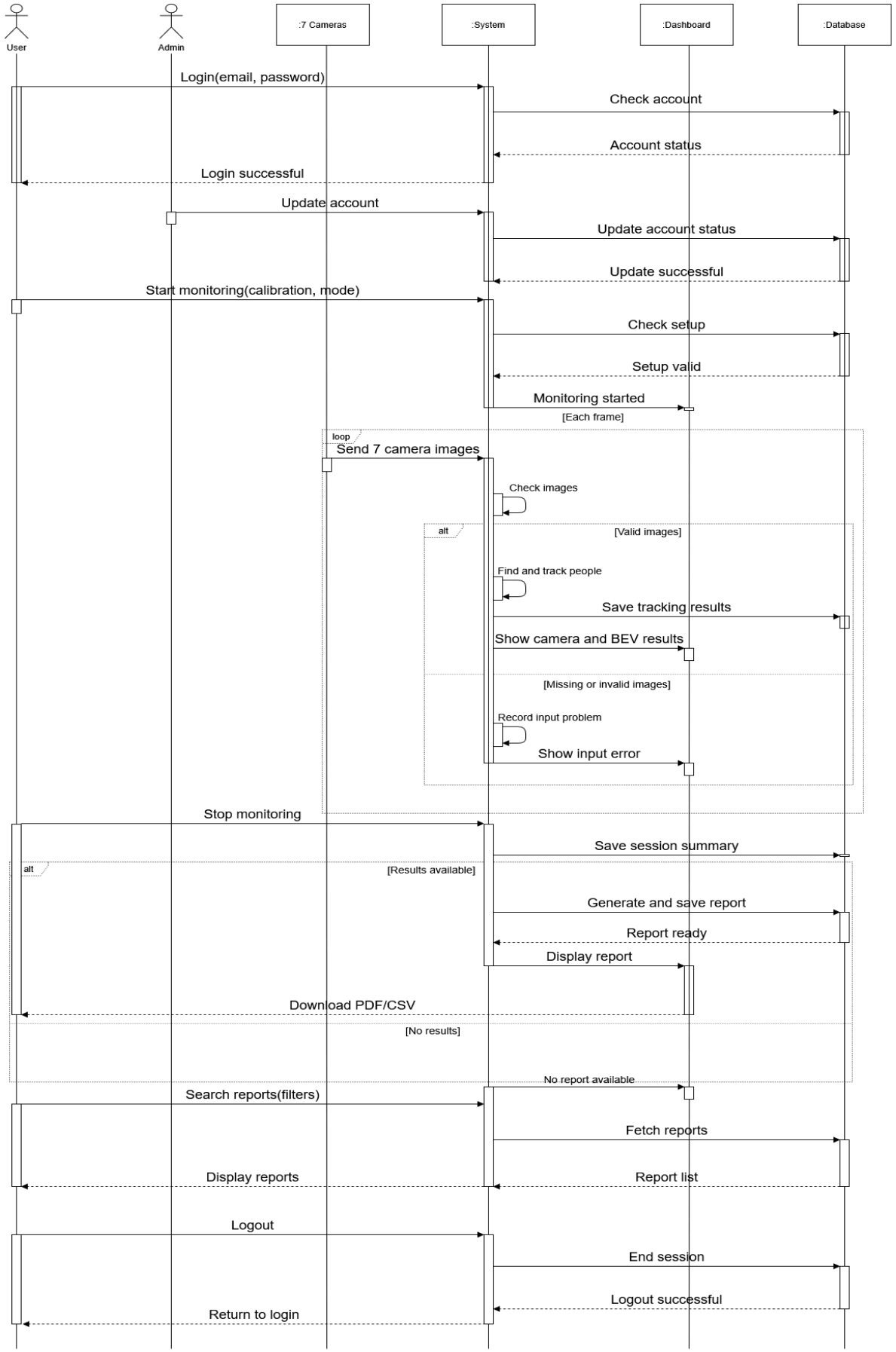


Figure 12 Sequence Diagram of the Retrack System

14. Data Flow Diagram

14.1 DFD-Level 0

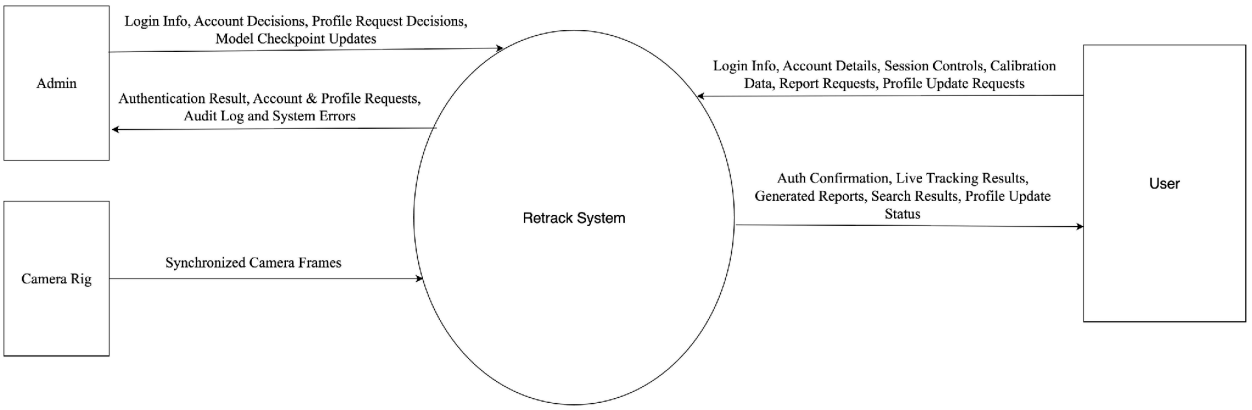


Figure 13 Data Flow Diagram – Level 0

14.2 DFD-Level 1

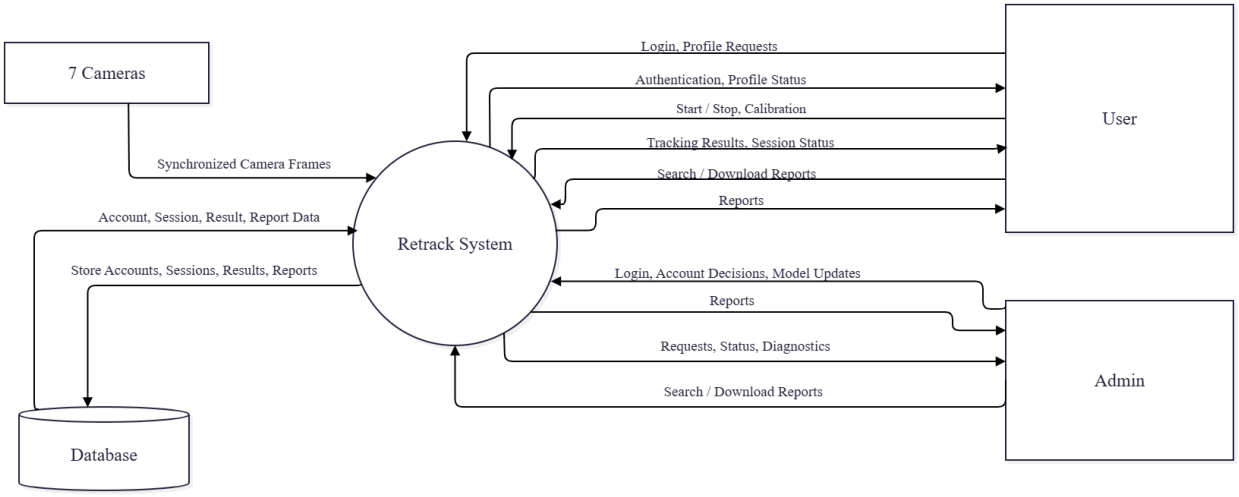


Figure 14 Data Flow Diagram – Level 1

15. JSON Tree Diagram

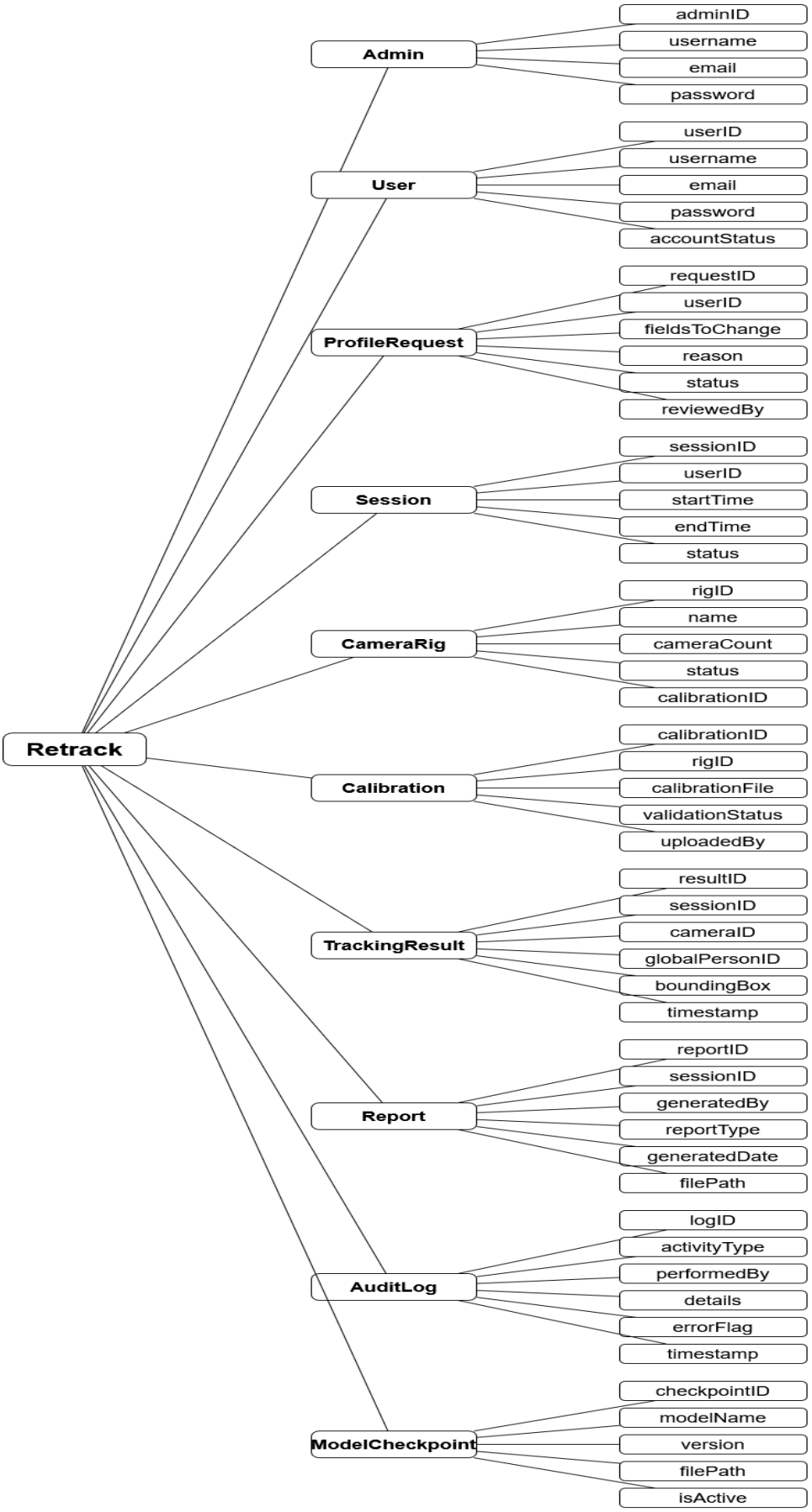


Figure 15 JSON Tree Diagram of the Tracking Result Structure

Dataset

We will evaluate our work on Wildtrack [3] which has become a standard benchmark for multi-view pedestrian detection and tracking systems. Its wide adoption allow us to directly compare our results with prior work. It has multiple advantages over synthetic datasets including captured in real world, unscripted public setting, dense crowds with occlusion. The dataset was recorded using seven synchronized and calibrated cameras with overlapping fields of view, covering an area of 12 m × 36 m. Each camera captures frames at a resolution of 1920 × 1080, and 400 frames are annotated at 2 fps. The ground plane is discretized into a 480 × 1440 grid, where each cell corresponds to a 2.5 cm × 2.5 cm region. On average, each frame contains about 20 pedestrians, and each location is visible from approximately 3.74 cameras. Following standard practice [1, 2, 4] we use the first 360 frames for training and the remaining 40 for testing.

16. References

[1] T. Teepe, P. Wolters, J. Gilg, F. Herzog, and G. Rigoll, “EarlyBird: Early-fusion for multi-view tracking in the bird’s eye view,” in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, 2024.

[2] T. Teepe, P. Wolters, J. Gilg, F. Herzog, and G. Rigoll, “Lifting multi-view detection and tracking to the bird’s eye view,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2024.

[3] T. Chavdarova *et al.*, “WILDTRACK: A multi-camera HD dataset for dense unscripted pedestrian detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018.

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[6] A. W. Harley, Z. Fang, J. Li, R. Ambrus, and K. Fragkiadaki, “Simple-BEV: What really matters for multi-sensor BEV perception?” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023.